AYMANE AIT DADS

Software Dev Engineer Intern - Machine Learning | ML Pipeline Engineering · LoRA Fine-Tuning · Scalable ML Systems

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, $106K+ prize pool) by engineering an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer with BF16 precision and 8,192-token context. Reduced manual network analysis time by 60% at Orange Maroc by deploying Random Forest and K-Means models on 100K+ production records, delivering output directly adopted by the network optimization team. Brings hands-on experience in PyTorch, Hugging Face Transformers, PEFT, and scalable ML pipeline design - skills that map directly to Amazon Robotics' need for engineers who build reliable, high-impact ML systems. Aims to apply rigorous ablation methodology and production-grade ML engineering to Amazon Robotics' real-world automation challenges at scale.

CORE COMPETENCIES

Machine Learning Engineering | Scalable ML Systems | Model Fine-Tuning | Python Development | Data Pipeline Design | Computer Vision | NLP & Reasoning | Collaborative Team Environment

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a **Random Forest + K-Means clustering pipeline** on 100K+ fiber-optic network records - upload speed, latency, jitter - to classify network quality across thousands of nodes.
- Designed a customer performance profiling system mapping network behavior to **5 commercial service tiers**, with output adopted directly by the network optimization team for operational decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible data processing scripts in Python and Pandas.
- Delivered stakeholder presentations translating **model outputs into maintenance recommendations**, bridging the gap between data science results and business action using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress)
Kaggle Competition · June 2026 · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, alpha=32, BF16, completion-only loss, 8,192-token context, grad_accum=32) targeting all attention projections, MLP layers, and lm_head for full-model adaptation.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and an 8-bit operation solver generating 64 verified CoT rows for hard-task training.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection - NTIRE 2026 @ CVPR
EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in EURECOM Class

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head using dual learning rates - 2e-6 for the backbone, 5e-4 for the head - trained on 250K images across 25+ generator types; also submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark.
- Discovered that **1-epoch training achieved 0.793 AUC** versus 0.767 AUC at 4 epochs, confirming superior out-of-distribution generalization; applied a 3-model ensemble with 10-view TTA (flips, crops, blur, resizes) weighted by validation AUC to maximize leaderboard score.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · AdamW · Kaggle

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM
Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Statistics, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya
2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Hugging Face Transformers | Unsloth | TRL | PEFT | Scikit-learn | vLLM

LLM Fine-Tuning & NLP: LoRA | QLoRA | SFTTrainer | Completion-only Loss | BF16/FP16 Mixed Precision | Chain-of-Thought | Prompt Engineering

Data Science & MLOps: Python | Pandas | NumPy | SQL | Git | Docker | Power BI | Feature Engineering